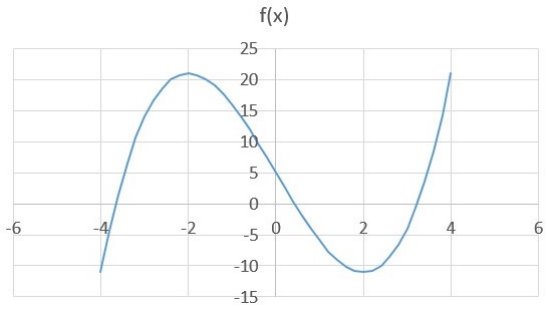
COMP1046 Part 2: Tutorial 7: Optimization Exercises

1. Consider the function $f(x) = x^3 - 12x + 5$.

   (a) Find all critical numbers for f.
   ANSWER:
   The function is well-defined for all real numbers x, so look for cases where $f'(x) = 0$.
   $$f'(x) = 3x^2 - 12 = 0.$$
   This is solved for $x = -2$ and $x = 2$ so these are the critical numbers.

   (b) Sketch the graph of f to determine whether each critical number is a minima, maxima or neither.
   ANSWER:



   Therefore $x = -2$ is a maxima and $x = 2$ is a minima.

2. Find two real numbers that sum to 100 and whose product is a maximum.
   ANSWER:
   Let x, y be the two real numbers. Then $x + y = 100$ and maximize xy.
   Find critical numbers,
   $$\frac{d}{dx}xy = \frac{d}{dx}x(100 - x) = 100 - 2x = 0.$$
   Hence $x = 50$ and $y = 50$.
   This is a maximum since any deviation $\delta \neq 0$ from the solution gives a smaller product:
   $$(x + \delta)(y - \delta) = xy + (y - x)\delta - \delta^2 = xy - \delta^2 < xy.$$

3. A vase manufacturer need a box-shaped container to carry the vases on their truck. The container has to be 1 meter wide to fit on the truck and needs to be $3m^3$ in volume. The roofing material costs 50 CNY per m^2, the side material costs 40 CNY per m^2 and the floor material costs 10 CNY per m^2. The *standard* container has length 3 meters and height 1 meter. However, the analytics team figure out how to build the container at minimum cost.

   (a) How much will the standard container cost to build?

   (b) What are the dimensions of the container that gives minimum cost?

   (c) How much saving does the new container provide?

   ANSWER:
   If x is the length and h the height of the container, the volume is given by $xh = 3$ since the width is 1 meter.
   The cost is given by

$$C = 50x + 40 \times (2xh + 2h) + 10x = 20(3x + 4xh + 4h).$$

   (a) Therefore, the standard container has $x = 3$ and $h = 1 \Rightarrow C = 20(9{+}12{+}4) =$ 500 CNY.

   (b) What are the dimensions of the container that gives minimum cost? Substitute volume equation $xh = 3$ into the cost equation to get it just in terms of h:

$$C = 20(9/h + 12 + 4h).$$

   Differentiate C to find extemum, using the result from part 2:

$$\frac{dC}{dh} = 20(-9/h^2 + 4) = 0$$

$$\Rightarrow h^2 = \frac{9}{4} \Rightarrow h = \frac{3}{2}$$

   since h must be a positive number and cannot be zero since then the volume would be zero. Then $x = 3/h = 2$.
   So, overall, the optimal container has height 1.5 meters and length 2 meters.

   (c) Cost with this design is $C = 20(6 + 12 + 6) = 480$ CNY.
   Therefore the saving is 20 CNY.

4. Consider designing a particular hard disk. This hard disk can run up to speeds where a block of data can be read in 4ms. So time to read a block is $t > 4$. However, the faster that the hard disk runs, the more prone to error in reading a block, and the probability of error is given by

$$p = \frac{1}{1 + \exp(t/2 - 3.2)}.$$

If an error occurs, it needs to be read again and this takes t ms. If this also fails with an error, then an attempt to read the block is made again, and so on, until the block is read without error. We want to set the time t in order to minimize the *expected value* of time to read without error.

(a) Show that the expected value of time to read without error, as a function of t, taking into account the probability of error, is

$$T_e(t) = t\left(\frac{1}{1-p}\right).$$

Hint: you will need the formula for an infinite geometric progression: for $0 \leq r < 1$,

$$1 + r + r^2 + r^3 + \cdots = \sum_{i=1}^{\infty} r^i = \frac{1}{1-r}.$$

ANSWER:

- The expected time is the time for the first read plus probability of error times time for next read and so on:

$$\begin{aligned} T_e(t) &= t + p(t + p(t + p(t + \cdots))) \\ &= t + pt + p^2t + p^3t + \cdots \\ &= t\left(\tfrac{1}{1-p}\right) \end{aligned}$$

- Alternative answer (following class discussions):
Consider this sequence of independent events leading to successful block read:

Event	Total time taken	Probability
No errors	t	$(1-p)$
1 error	$2t$	$p(1-p)$
2 errors	$3t$	$p^2(1-p)$
$\cdots$	$\cdots$	$\cdots$
i errors	$(i+1)t$	$\mathrm{p}^i(1-p)$
$\cdots$	$\cdots$	$\cdots$

Then expected value of time taken to success read is given by

$$\begin{aligned} T_e(t) &= \textstyle\sum_{i=0}^{\infty} (i+1)t\, p^i(1-p) \\ &= \textstyle\sum_{i=0}^{\infty} (i+1)tp^i - \sum_{i=0}^{\infty} (i+1)tp^{i+1} \end{aligned}$$

Notice that taking $j = i + 1$, the second sum can be rewritten as

$$\sum_{j=1}^{\infty} jtp^j = \sum_{j=0}^{\infty} jtp^j$$

since the term $jtp^j = 0$ when $j = 0$. Then,

$$\begin{aligned} T_e(t) &= \textstyle\sum_{i=0}^{\infty} (i+1)tp^i - \sum_{j=0}^{\infty} jtp^j \\ &= \textstyle\sum_{i=0}^{\infty} tp^i \\ &= t\left(\frac{1}{1-p}\right) \end{aligned}$$

(b) Compute the gradient

$$\frac{d}{dt} T_e(t).$$

Hint: $\frac{d}{dt}\exp(t) = \exp(t)$.

ANSWER:
Substituting the formula for p,

$$\begin{aligned} T_e(t) &= t\left(\frac{1}{1-\frac{1}{1+\exp(t/2-3.2)}}\right) \\ &= t\left(\frac{1+\exp(t/2-3.2)}{1+\exp(t/2-3.2)-1}\right) \\ &= t\left(\frac{1+\exp(t/2-3.2)}{\exp(t/2-3.2)}\right) \\ &= t\left(\frac{1}{\exp(t/2-3.2)}+1\right) \end{aligned}$$

Then, using the product rule and then the chain rule,

$$\begin{aligned} \tfrac{d}{dt}T_e(t) &= \tfrac{d}{dt}t\left(\tfrac{1}{\exp(t/2-3.2)}+1\right) + t\tfrac{d}{dt}\left(\tfrac{1}{\exp(t/2-3.2)}+1\right) \\ &= \left(\tfrac{1}{\exp(t/2-3.2)}+1\right) + t\tfrac{-1}{\exp(t/2-3.2)^2}\exp(t/2-3.2)\times\tfrac{1}{2} \\ &= \left(\tfrac{1}{\exp(t/2-3.2)}+1\right) - \tfrac{t/2}{\exp(t/2-3.2)} \\ &= \tfrac{1-t/2}{\exp(t/2-3.2)}+1 \end{aligned}$$

(c) Compute the first 10 iterations of gradient descent to approximate the value of t that minimized $T_e(t)$.

Hint: Use Python to implement gradient descent. Use math.exp for the exponent function.
ANSWER:
See the separate Python notebook file, available with these answers.
Approximate solution is $t = 8.868$ ms, which gives $T_e(t) = 11.45$ ms.

(d) Based on your gradient descent, what is the minimum value of $T_e(t)$?
ANSWER:
$T_e(8.868) = 11.45$ ms.

5. Re-use your Python code to impement gradient descent for Highway Building problem in Lecture 12 slides 17 to 21. Ensure you get the same solution.

ANSWER:
See the separate Python notebook file, available with these answers.